A regional paleomagnetic study of Archean rocks from the Superior Geotraverse area, northwestern Ontario[1]

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Preliminary paleomagnetic data are reported for Archean metavolcanics, felsic and mafic intrusives, gneisses, and iron formations from the Quetico, Shebandowan, and Wabigoon belts in the western Superior Structural Province. Eleven of the 23 formations sampled have been studied in detail using stepwise alternating-field demagnetization and, in some cases, thermal demagnetization. Two characteristic components of magnetization are revealed. One is widespread in occurrence and reasonably well grouped ($D = 4.5°$, $I = +55.7°$, $k = 17.9$, $\alpha_{95} = 5.4°$, $N = 40$ samples for the Shelley Lake granite for example). It is a regional magnetic overprint due to the Kenoran orogeny, acquired, according to the position of its paleopole (71.7°E, 77.3°N, $\delta p = 5.5°$, $\delta m = 8°$ for the Shelley Lake granite) on the Laurentian apparent polar wander path and to independent radiometric evidence, about −2600 Ma. The second magnetic component is spotty in occurrence, more prevalent in the Wabigoon belt than elsewhere, and everywhere rather scattered. The only fully reliable determination is $D = 66.0°$, $I = -5.1°$ ($k = 10.2$, $\alpha_{95} = 15°$, $N = 11$ samples) for the Wabigoon gabbro. The corresponding paleopole is either 17.8°E, 13.1°N ($\delta p = 7.5°$, $\delta m = 15°$) falling at about −2800 Ma on the apparent polar wander path or the antipole of this direction, dating between −1350 and −1250 Ma. The younger date, implying a late Proterozoic metamorphic event in the region, is more likely than the older age, which would require that the magnetization be primary or a pre-Kenoran overprint. The metamorphism seems to have occurred too early to have been caused by igneous activity ca. 1100 Ma in the nearby Keweenawan basin. About −2600 Ma, the region was in its present orientation, but at a latitude of 20 to 35°N and probably drifting northward. About −1250 Ma, the region was equatorial but rotated 90°, so that presently east–west trending greenstone belts were north–south. It is possible, but unproven, that a paleoocean existed between the Wabigoon subprovince and the other belts about −1250 Ma, the Wabigoon region having since moved >500 km westward (present-day direction) to its present location.

On rapporte les données préliminaires sur le paléomagnétisme des roches métavolcaniques, les intrusions felsiques et mafiques, les gneiss et les formations de fer de l'Archéen dans les zones de Quetico, de Shebandowan et de Wabigoon dans la partie ouest de la province structurale du Lac Supérieur. On a étudié en détail 11 des 23 formations échantillonnées en procédant à une désaimantation dans un champ alternatif et, dans certains cas, à une désaimantation thermique. On a mis en relief deux composantes caractéristiques de désaimantation. Une est d'occurrence générale et forme un faisceau raisonnablement bien défini ($D = 4.5°$, $I = +55.7°$, $k = 17.9$, $\alpha_{95} = 5.4°$, $N = 40$ échantillons pour le granite de Shelley Lake, par exemple). C'est une surimpression magnétique régionale causée par l'orogénèse kénoréenne, acquise il y a environ 2600 Ma, selon la position de son paléopôle (71.7°E, 77.3°N, $\delta p = 5.5°$, $\delta m = 8°$ pour le granite de Shelley Lake) sur le trajet du cheminement polaire apparent laurentien et selon des preuves radiométriques indépendantes. La seconde composante magnétique est plus irrégulière, elle se retrouve surtout dans la zone de Wabigoon et elle est partout plutôt diffuse. La seule détermination fiable indique $D = 66.0°$, $I = -5.1°$ ($k = 10.2$, $\alpha_{95} = 15°$, $N = 11$ échantillons) pour le gabbro de Wabigoon. Le paléopôle correspondant est soit à 17.8°E, 13.1°N ($\delta p = 7.5°$, $\delta m = 15°$) correspondant à environ 2800 Ma sur le trajet du cheminement polaire apparent, soit l'antipôle de cette direction, dont l'âge est compris entre 1350 et 1250 Ma. La date plus récente, impliquant un événement métamorphique à la fin du Protérozoïque dans la région semble plus plausible que la date plus ancienne qui exigerait que l'aimantation soit primaire ou une surimpression pré-kénoréenne. Le métamorphisme semble s'être produit trop tôt pour avoir été causé par l'activité ignée, autour de 1100 Ma, dans le bassin keweenawien adjacent. Il y a environ 2600 Ma, la région était dans son orientation présente, mais à une latitude de 20 à 35°N et probablement dérivant vers le nord. Il y a environ 1250 Ma, la région était équatoriale mais avait subi une rotation de 90° de sorte que les zones de roches vertes présentement est–ouest étaient nord–sud. Il est possible, mais non démontré, qu'un paléo-océan ait existé entre la sous-province de Wabigoon et les autres zones il y a environ 1250 Ma, et que depuis ce temps, la région de Wabigoon se soit déplacée >500 km vers l'ouest (direction actuelle) jusqu'à sa position présente.

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Introduction

Previous paleomagnetic studies of North American Archean rocks (e.g., Strangway 1964; Fahrig *et al.* 1965; Bergh 1970; Irving and Naldrett 1977) have established the reality of apparent polar wander (i.e., continental drift) between −2.8 and −2.5 Ga. The purpose of the present paper is not primarily to add further poles to the Archean apparent polar wander path (APWP)—Track 6 of Irving and Naldrett (1977). It is rather to combine present knowledge, albeit incomplete, of the Precambrian APWP for North America with preliminary paleomagnetic results from a large number of formations in a tentative *regional* interpretation of the magnetic, metamorphic, and tectonic history of the southern Superior Geotraverse area.

This area of the Superior Structural Province has been regionally metamorphosed to greenschist facies (increasing to lower amphibolite facies in the English River gneiss belt to the north), as a result of the Kenoran orogeny, which culminated about −2.6 Ga. Later thermal events that may have affected the area are the −1.1 Ga volcanism and intrusive activity in the Keweenawan basin to the southeast, and the ca. −1.25 Ga Mackenzie igneous episode. No Mackenzie-age dikes have been identified in the geotraverse area, but igneous activity of this age is widespread in the Superior Province and may have resulted in mild regional heating of the area.

At the outset, it is well to be clear as to the possibilities and the limitations of the paleomagnetic method applied to a region rather than to a single formation. Firstly, we gain insight into the response of magnetic minerals to regional metamorphism, particularly if there are variations in metamorphic grade through the region. A knowledge of which rock types are likely to have the strongest and most enduring induced and remanent magnetizations under specific metamorphic conditions is helpful in interpreting magnetic anomalies and in selecting promising targets for detailed paleomagnetic studies.

Secondly, we may succeed, in studies of dated individual units, in strengthening poorly defined parts of the APWP, Track 6 for instance. The difficulties are considerable for units as old as Archean. Their natural remanent magnetizations (NRM's) are generally multicomponent: vector sums of thermal and chemical magnetizations acquired at different stages of a turbulent history. Unravelling the "overprinted" NRM components from one another and from the primary component, if any survives, is a major challenge (see, e.g., Roy and Lapointe 1978), nor is it necessarily easy to decide to which component a given radiometric date refers.

However, even if a specific NRM component cannot be dated unambiguously or if the unit is not dated at all, the North American APWP is now known sufficiently well in broad outline to bracket the possible ages of the components, and of the metamorphic events that produced them, in one or a few narrow intervals. In regions of few radiometric age determinations, magnetic ages are valuable clues to metamorphic history.

Finally, the directions of the various NRM components provide spot readings of paleolatitude and paleoazimuth of the continent or protocontinent at times of active metamorphism. As an example, consistent differences in directions and (or) blocking temperatures of a particular NRM component, within a single metamorphic belt or subprovince, may reflect age differences between units, uplift at different times, or erosion to different depths.

In fortunate circumstances, one can hope to test whether global tectonics operated on a different scale in the Archean, and perhaps even whether the tectonic style was other than one of rigid plates or platelets. To be specific, the fortunate circumstances needed to test for a paleoocean between presently adjacent structural belts are opening and closing in a north–south direction (since longitude is not recorded paleomagnetically) of several degrees of latitude, preferably near the paleoequator, where magnetization direction changes much more rapidly with latitude than near the poles. Relative drift of less than a few hundred kilometres is not likely to be detectable under any circumstances, but relative rotations of even a few degrees are detectable magnetically, especially if they occurred at low paleolatitudes. Once sufficient data become available, the same reasoning can be extended to continent-wide and global reconstructions in the early Precambrian.

The present study is no more than a first step in the directions sketched above. With specific exceptions, the individual paleomagnetic directions reported here are not yet of a quality to stand on their own. They are to be regarded as the early pieces in a puzzle, which taken together hint at regional trends.

Regional Magnetic Results

Samples oriented by sun and (or) magnetic compass were collected during four field seasons at the 120 sites shown in Fig. 1. Results are tabulated for 23 units or groups of related units in Table 1. Sus-

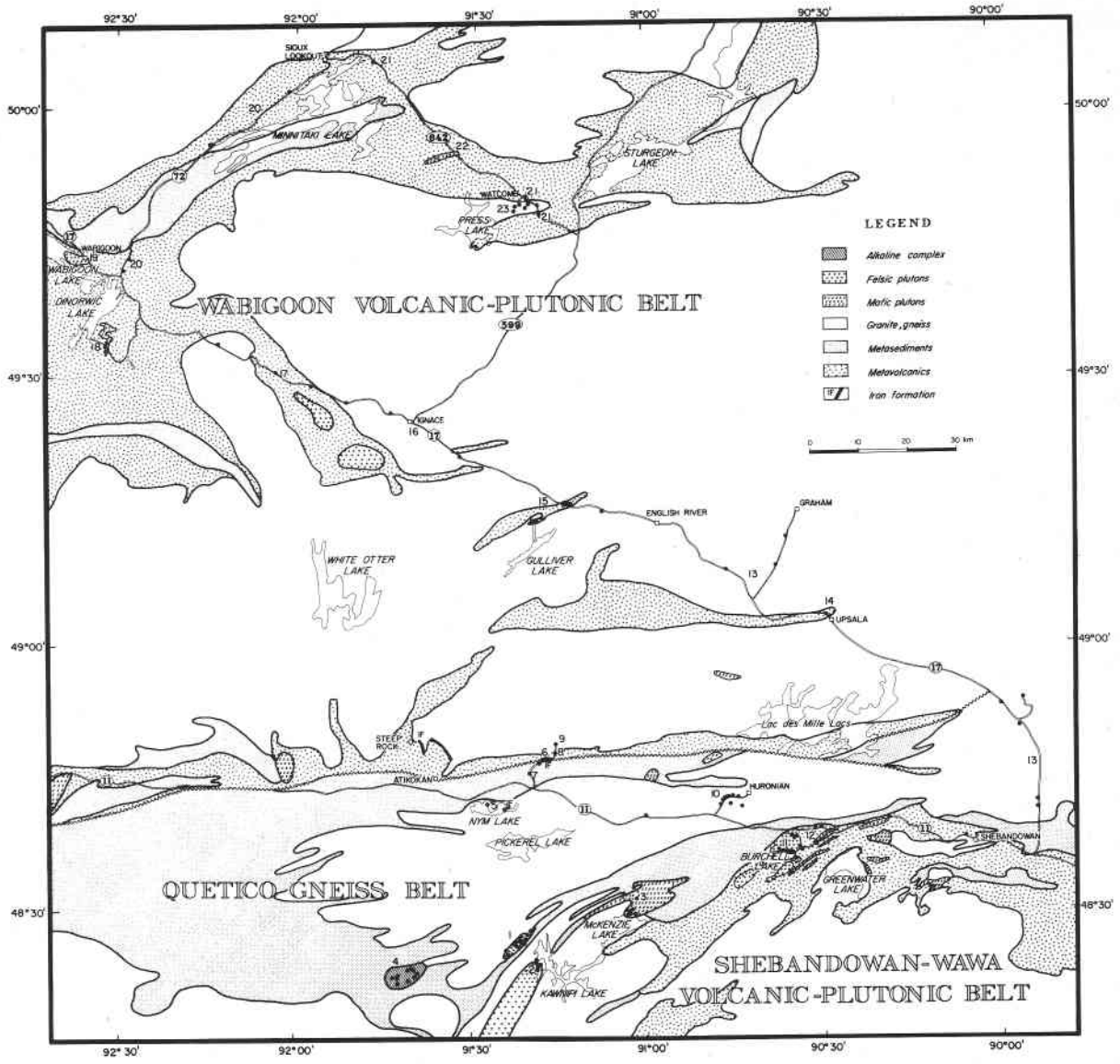


FIG. 1. Sketch map of the southern geotraverse area, showing sampling sites (dots). Formations or groups of related sites are numbered to correspond with Table 1.

ceptibility was measured with an alternating current (AC) bridge and NRM's with a Digico spinner magnetometer capable of resolving intensities $\geq 10^{-7}$ emu cm^{-3}.

In detailed paleomagnetic studies, three specimens were "cleaned" by three-axis stepwise AF (alternating field) demagnetization to 1000 Oe (7.95 $\times$ 10^4 A/m) peak fields. Companion thermal cleaning is underway on one specimen of selected samples from formations 1, 2, 3, 6, 10, 11, and 12. Thermal work will eventually be extended to all stable formations. Since magnetization generally falls to a few percent of its initial value during cleaning, formations with NRM's largely $\geq 10^{-5}$ emu cm^{-3} were most profitably studied in detail. However, high NRM intensity is no guarantee of stable behaviour, nor does low average intensity invariably disqualify a unit from detailed study (e.g., units 10 and 23).

Natural remanent magnetization intensities varied over many orders of magnitude among rock types, among formations of similar rock type (Fig. 2), and among samples of individual formations. Susceptibility was much less variable, particularly within units. Natural remanent magnetization variability, therefore, seems to express primarily

wide variability in the stability over time of the "original" (not necessarily primary) NRM and not variations in magnetic mineral content. Decay of NRM, whether spontaneous or due to thermal or chemical overprinting, in turn reflects grain size of the magnetic minerals or effective size of exsolved and subdivided grains. Rather small shifts in mean grain size spell the difference between paleomagnetic stability and instability.

If regional trends or consistent differences between structural subprovinces exist in this area, they have no obvious expression in the magnetic results of Table 1. Felsic intrusives from the Quetico, Shebandowan, and Wabigoon belts had similar high induced and remanent magnetizations. Magnetic variation from site to site in individual greenstone belts was orders of magnitude greater than differences in $\bar{J}_{NRM}$ or $\bar{k}$ from one greenstone belt to another.

The regional grade of metamorphism in the southern geotraverse area is middle to upper greenschist. Even this relatively mild degree of metamorphism has profoundly affected the magnetic minerals. Basalts forming in the oceans today and fresh Tertiary basalts in general have magnetite or titanomagnetite contents of 1% or so, induced magnetizations of $\sim 10^{-3}$ emu cm^{-3}, and Q_n's > 1 (often $\gg 1$). Their NRM's tend to be about 10 times those of related mafic intrusive rocks. The metavolcanic rocks in Table 1 have 10^{-6} emu cm$^{-3} \leq \bar{k} \leq 10^{-4}$ emu cm^{-3}: they contain 10 to 1000 times less magnetite than their modern-day equivalents. On average, their Q_n's are very low, indicating a coarse grain size. Greenschist-facies metamorphism has been responsible for destroying the finer-grained, high-stability fraction of the magnetic minerals in the volcanics. Mafic intrusives have fared somewhat better, presumably because of their larger initial grain sizes. Their $\bar{k}$ and $\bar{J}_{NRM}$ values are about twice those of the volcanics.

It is more difficult to say what magnetic minerals may have been *added* to these rocks in the course of metamorphism. The felsic intrusives as a class are remarkably magnetic (Table 1), although the sampling procedure of course has the bias that the more magnetic formations are sampled in more detail. Certainly from the point of view of anomaly interpretation, felsic plutons in this area are more likely to have aeromagnetic expression than mafic ones. Greenstone belts can almost be ignored except for their associated iron formations. Felsic intrusions often have Q_n fairly close to 1, in contrast to the mafic rocks. The remanent component cannot, therefore, be ignored in interpreting anomalies. The same is true for the iron formations.

The felsic intrusions also turned out to be more promising candidates for paleomagnetic study than other rock types, as the following sections will show.

Stable Remanent Magnetizations

The NRM's of three specimens of each sample in Table 1 were AF cleaned in 11 steps to 1000 Oe (7.95×10^4 A/m). Twelve formations were rejected because of the high proportion of magnetizations that were either unmeasurably weak, directionally unstable during cleaning, or directed near the local geomagnetic field direction ($D = 4°$E, $I = +70°$) and of probable recent viscous origin.

The remaining 11 formations were sampled in detail in the third and fourth field seasons. They too have their share of weak, soft, viscous, or unstable samples—not surprising in view of the generally low Q_n's. The stably magnetized samples contain, in addition to a possible viscous remanence erased at low AF's, one or both of the characteristic stable paleoremanences whose directions are plotted in Figs. 3 and 4. Group 1 directions tend to have north (or south) declinations and intermediate (40–65°) inclinations. Group 2 directions tend to have shallow inclinations with east to northeast (or west to southwest) declinations. Reversals of the paleofield are recorded in both groups. The proportion of normal and reversed samples is variable and not close to 50:50, suggesting infrequent reversals.

A few formations show only group 1 or only group 2 NRM's, but most have examples of both, frequently superimposed in individual samples. Superimposed (or multicomponent) NRM's were separable in two cases: (1) if one component was completely demagnetized at high AF steps, so that the remanence vector reached a stable endpoint direction or (2) if one component was not demagnetized at all over some AF range, usually at low AF steps, so that the direction of the other component could be deduced by the Zijderveld diagram vector-subtraction technique (Zijderveld 1967). Otherwise, although the magnetization vector followed a "remagnetization circle" (Halls 1976) joining groups 1 and 2, neither component could be isolated from the data and used in defining the mean group directions.

Sample mean directions of stable or isolated NRM's are plotted in Figs. 3 and 4. Each is the average of three specimen directions at the AF step where these directions grouped best. If specimen directions diverged $\geq 20°$, the sample was rejected, even though the specimen results were often individually stable. (Such results usually indicated lightning strikes and had anomalous directions as well as high between-specimen dispersion.)

TABLE 1. Natural remanent magnetization and susceptibility data and paleomagnetic stability of formations sampled in the Superior Geotraverse area

Formation or area sampled	Tectonic belt	No. of sites	No. of samples	$\bar{J}_{NRM} \times 10^6$ (emu/cm^3)	$\bar{k} \times 10^6$ (emu/cm^3 Oe)	$\bar{Q}_n$	Stability
(1) Shelley Lake granite	Quetico	12	55	425	406	2.20	Mostly stable
(2) Kawnipi Lake granite	Quetico	5	12	19.0	(13.7)	(2.22)	Mostly unstable
(3) McKenzie Lake diorite, volcanics	Shebandowan	8	19	2.0	63.3	0.16	Intermediate
(4) Poohbah Lake alkaline complex	Quetico	10	36	210	82.8	2.87	Stable
(5) Nym Lake syenite	Quetico	2	6	0.46	(1.7)	(0.54*)	Unstable
(6) Sapawe iron formation	Wabigoon	5	26	770	924	1.23	Intermediate
(7) Sapawe Road granite, syenite	Quetico	2	5	Granite: 260 Syenite: 0.26	(696)	(0.92)	Unstable
(8) Sapawe Road volcanics, gabbro	Wabigoon	2	9	10.7	(43.6)	(0.49*)	Intermediate
(9) Wabash Lake gneiss	Wabigoon	1	4	0.28			Unstable
(10) Huronian granite, paragneiss	Quetico	9	35	3.3	19.9	1.57	Mostly stable
(11) Burchell Lake granite	Shebandowan	11	53	49	172	0.77	Mostly stable
(12) Shebandowan gabbros, volcanics	Shebandowan	7	27	7.9	161	(0.27)	Intermediate
(13) Hwy. 11 to Graham gneisses, granites	Wabigoon	7	20	7.4	168	(0.53)	Mostly unstable
(14) Upsala greenstone	Wabigoon	1	5	66	(42.5)	(3.1*)	Unstable
(15) Gulliver greenstone	Wabigoon	6	25	27	87.4	0.62*	Intermediate
(16) English River to Ignace, gneisses	Wabigoon	7	29	4.7	93.6	(0.34)	Mostly unstable
(17) Raleigh Lake greenstone	Wabigoon	2	6	1.5	(60.8)	(0.05*)	Unstable
(18) Dinorwic gabbro	Wabigoon	1	3	155	(48.8)	(6.35*)	Mostly unstable
(19) Wabigoon gabbro	Wabigoon	8	32	13.0	177	0.15*	Mostly stable
(20) Hwy. 72, greenstones	Wabigoon	3	4	3.7			Unstable
(21) Hwy. 642, volcanics, gabbro	Wabigoon	4	11	0.28	(48.8)	(0.01*)	Unstable

TABLE 1. (*Concluded*)

Formation or area sampled	Tectonic belt	No. of sites	No. of samples	$\bar{J}_{NRM} \times 10^6$ (emu/cm^3)	$\bar{k} \times 10^6$ (emu/cm^3 Oe)	$\bar{Q}_n$	Stability
(22) Yonde granite	Wabigoon	1	4	13.3			Intermediate
(23) Watcomb granite	Wabigoon	6	25	1.7	19.8	0.17*	Intermediate
All felsic intrusives		57	223	46.5	139	0.67*	
All gneisses		16	62	2.85	(92)	(0.06*)	
All mafic intrusives		21	67	13.3	120	0.22*	
All volcanics		21	73	5.15	69	0.15*	
Iron formations		5	26	770	(1000)	(1.54*)	
All formations		120	451	(23)			

NOTES: Formation number is keyed to Fig. 1. Q_n is the Koenigsberger ratio, J_{NRM}/kH_p, where k is susceptibility and H_p is the intensity of the present Earth's field, nominally 0.5 Oe (39.8 A/m). $\bar{J}_{NRM}$, $\bar{k}$, and $\bar{Q}_n$ figures tabulated are geometric mean values. There were 1267 individual determinations of J_{NRM} but only 156 measurements of k and Q_n. $\bar{J}_{NRM}$, $\bar{k}$, and $\bar{Q}_n$ values in parentheses are possibly unrepresentative, usually because the number of measurements was small. $\bar{Q}_n$ values with an asterisk indicate that $\bar{J}_{NRM}/0.5\ \bar{k}$ is tabulated as being more representative than the mean of the measured Q_n's.

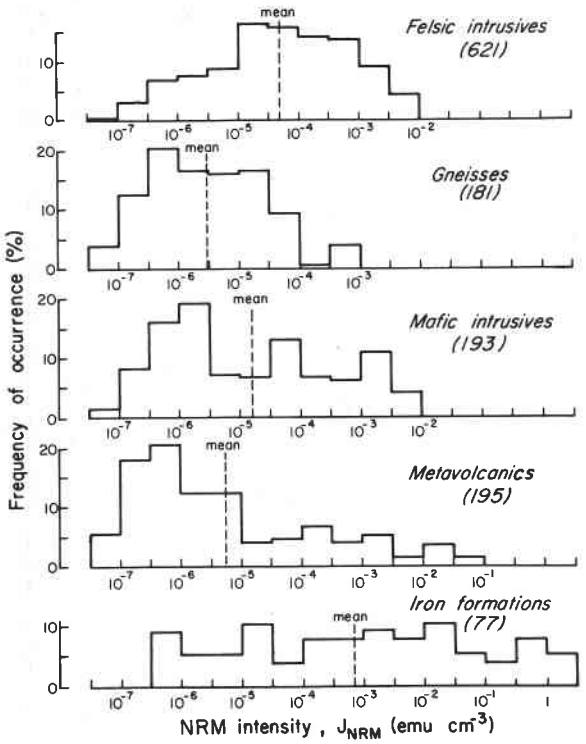


FIG. 2. Normalized histograms of NRM intensity by rock type. Total numbers of measurements are shown in parentheses.

It is obvious to the eye that the directional scatter in Figs. 3 and 4 varies from unit to unit and is greater for group 2 than for group 1 results. Mean magnetization directions (unit weight to samples) and Fisher statistical parameters κ and α_{95} are given in Tables 2 and 3. κ measures dispersion: for acceptable precision, $\kappa > 10$. α_{95} measures probability that the mean is correctly located: we can be 95% confident the true mean is within α_{95} degrees of the direction tabulated. Ideally, $\alpha_{95} < 10°$, but $10° < \alpha_{95} < 20°$ may be acceptable if the precision κ is adequate, since α_{95} (unlike κ) decreases as the number of samples (N) increases.

There is a further criterion for judging the results. For the sake of uniformity, unit weight has been given sample means for all units. It is more usual in paleomagnetism to carry out a 3-tier analysis, in which specimens, samples, and sites in turn are given unit weight. For several of the listings in Table 2, there are a sufficient number of sites with at least three group 1 sample means to allow a 3-tier analysis. In this first category are the Shelley Lake and Burchell Lake granites (Dunlop 1979) and the Poohbah Lake alkaline complex, all with $\kappa > 15$, $\alpha_{95} < 10°$ whether samples or sites are given unit weight.

Group 1 results from the Huronian and Watcomb granites and group 2 results from the Wabigoon gabbro fall in a second category. There are an adequate number of samples ($N > 10$) for a meaningful average but too few sites with $N \geq 3$ stable samples. Precision is acceptable ($\kappa \geq 10$) and $\alpha_{95} \leq 15°$. These results are preliminary but with further work will likely prove reliable.

All remaining results, including most group 2 results, fall in the category of inadequate. Some (Sapawe iron formation, group 1, and Burchell Lake granite, group 2) have unacceptable scatter ($\kappa < 10$, $\alpha_{95} \geq 20°$) and must be rejected out of hand. The others (McKenzie Lake diorite, group 1, and Shelley Lake granite, Shebandowan mafic rocks, Gulliver greenstone, and Ignace/Martin gneisses, group 2) have acceptable precision but too few

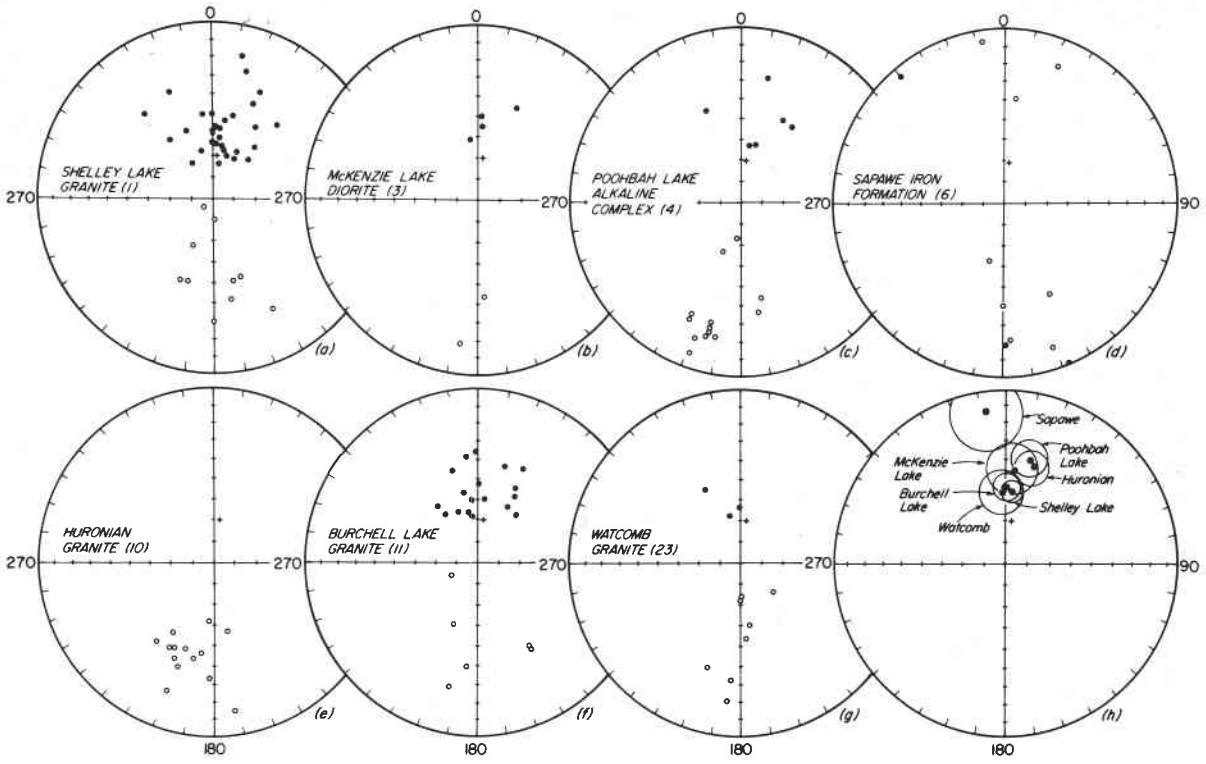


FIG. 3. (*a*)–(*g*) Sample mean group 1 magnetization directions after optimum AF cleaning for various formations. (*h*) Formation mean directions with circles of 95% confidence. A cross marks the present Earth's field direction.

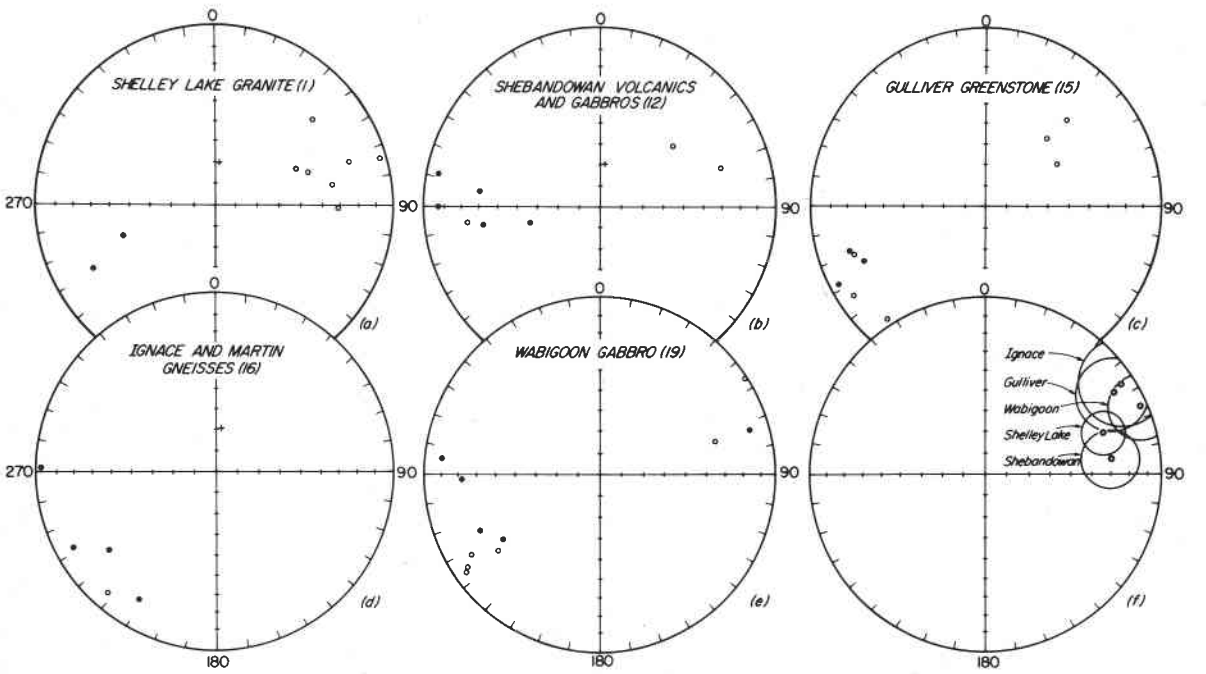


FIG. 4. (*a*)–(*e*) Sample mean group 2 magnetization directions after optimum AF cleaning. (*f*) Formation mean directions with circles of 95% confidence.

TABLE 2. Average group 1 paleomagnetic results from the Superior Geotraverse area

Formation	N	D	I	κ	α_{95}	Paleopole longitude, latitude	δp	δm
				Quetico gneiss belt				
(1) Shelley Lake granite	40	4.5°	+55.7°	17.9	5.4°	71.7E, 77.3N	5.5°	8°
(4) Poohbah Lake alkaline complex	19	12.3°	+39.2°	15.4	9.0°	63.5E, 62.0N	7°	11°
(10) Huronian granite, paragneiss	14	14.7°	+42.3°	25.8	7.9°	57.8E, 63.1N	6.5°	10°
				Shebandowan volcanic–plutonic belt				
(3) McKenzie Lake diorite, volcanics	6	4.6°	+45.5°	24.1	13.9°	77.7E, 68.1N	12°	18°
(11) Burchell Lake granite	24	0.0°	+54.3°	19.6	6.8°	89.4E, 76.2N	7°	10°
				Wabigoon volcanic–plutonic belt				
(6) Sapawe iron formation	11	351.5°	+13.4°	6.1	20.0°	100.7E, 47.4N	10.5°	20.5°
(23) Watcomb granite	11	356.7°	+55.6°	16.3	11.6°	99.5E, 76.1N	12°	16.5°

NOTES: N is number of samples averaged. (Each sample magnetization is the minimum dispersion average of three specimen magnetizations at some level of AF cleaning from 0–1000 Oe (0–7.95 × 10^4 A/m).) D, I are declination, inclination of magnetization. κ is Fisher's precision parameter; α_{95} is the radius of the cone of 95% confidence about the mean magnetization direction. δp, δm are the semiaxes of the oval of 95% confidence about the paleopole.

TABLE 3. Average group 2 paleomagnetic results from the Superior Geotraverse area

Formation	N	D	I	κ	α_{95}	Paleopole longitude, latitude	δp	δm
				Quetico gneiss belt				
(1) Shelley Lake granite	9	70.5°	−30.2°	21.2	11.4°	23.6E, 0.1N	7.5°	12.5°
				Shebandowan volcanic–plutonic belt				
(11) Burchell Lake granite	9	96.2°	−6.7°	5.7	22.0°	3.0W, 6.6S	11°	22°
(12) Shebandowan gabbros, volcanics	8	82.9°	−28.6°	13.7	15.4°	14.8E, 6.8S	10°	17°
				Wabigoon volcanic–plutonic belt				
(15) Gulliver greenstone	9	57.3°	−14.6°	8.6	18.5°	28.9E, 14.5N	10°	19°
(16) Ignace, Martin gneisses	5	55.9°	−8.7°	10.0	25.3°	28.2E, 17.8N	12.5°	25°
(19) Wabigoon gabbro	11	66.0°	−5.1°	10.2	15.0°	17.8E, 13.1N	7.5°	15°

NOTE: Symbols are as in Table 2.

samples ($N < 10$). Further work may confirm their reliability, although our attempts over four years to isolate a sufficient population of group 2 directions in *any* formation have proved discouraging.

The reason for plotting directions for all formations, including ones whose data barely pass, or in some cases fail, statistical tests, is our conviction that the results *in toto* have a story to tell, even though some results, taken individually, do not inspire confidence. One cannot use paleopoles for such weak results to better define the APWP, but it is fair to compare category 3 to other results to see if there are any obvious regional trends.

Note firstly that all group 1 means are distinct

from the present Earth's field direction in the area (i.e., the cross in Fig. 3*h* marking the present field does not fall within any of the 95% confidence circles). In all cases except the Watcomb granite, they are also distinct from the local dipole field direction ($D = 0°$, $I = +66.5°$), the long-term average about which the Earth's field fluctuates. The group 2 directions are unlike either the modern field or the group 1 directions. Thus, we are dealing with recordings of the ancient Earth's field imprinted at two quite different times.

At least a hint of a group 1 NRM is common to every rock type sampled (felsic and mafic intrusives, mafic and intermediate metavolcanics, paragneisses, and, more dubiously, iron formations) and to all three tectonic belts. Furthermore, the Burchell Lake, Shelley Lake, and Watcomb granites, from the Shebandowan, Quetico, and Wabigoon belts, respectively, have means that are not significantly different and are barely distinct from those of the Poohbah Lake complex and the Huronian granite and paragneiss. The times at which these various units formed are not known in any detail but they likely span a sufficient interval—100 to 200 Ma—for significant apparent polar wander to have occurred. On these grounds, the group 1 magnetization is unlikely to be primary. It represents a pervasic remagnetization or magnetic overprinting of regional extent.

The group 2 magnetization is also imprinted in many different rock types and there are examples from all three structural belts. However, many formations lack any trace of a group 2 component. It is rare in the Quetico belt but widespread in the Wabigoon volcanic subprovince. From this evidence, the group 2 NRM could be primary and confined to older units in all belts or it could be a late overprint due to a relatively mild metamorphic event that almost nowhere totally erased the group 1 NRM.

There is a suggestion in Fig. 4*f* that group 2 means from the Wabigoon belt form a cluster distinct from the single reliable Shebandowan belt direction. The one Quetico belt result is intermediate and, with present precision, not significantly different from either. These groupings will be shown later to be consistent with (although they do not prove) drift of the tectonic belts relative to one another.

Multicomponent Magnetizations and Relative Ages of NRM Components

One specimen of selected samples (28 from the Shelley Lake granite, 18 from the Burchell Lake granite, and nine from the Sapawe iron formation) have been thermally cleaned to 600°C. Thermal cleaning has three aims: (1) to identify the minerals carrying NRM and the NRM fraction carried by each; (2) possibly to reveal a new NRM component or components or to isolate superimposed components not separable by AF cleaning, and (3) to determine the relative ages of thermally overprinted NRM components.

This last deduction is possible because thermal demagnetization reproduces, in reverse and on a reduced time scale, the thermal remagnetizing process in nature. The first NRM component erased (the one with the lowest blocking temperatures) is the youngest, and the last to be erased is the oldest. Clearly this conclusion does not hold for chemical overprints, which can have blocking temperatures vastly in excess of temperatures experienced in nature during remagnetization. Thermal overprints, being partial TRM's (thermoremanent magnetizations), cannot have overlapping blocking temperature ranges (Thellier 1938). This constitutes a minimum test of thermal overprinting.

In the Shelley Lake granite, an average 35% of the NRM is carried by a mineral with Curie point between 250 and 300°C, 60% by a mineral with blocking temperatures from 300–580°C and a Curie point between 525 and 580°C, and 5% has blocking temperatures >600°C. In 20 thin sections examined in reflected light by L. D. Schutts, these minerals were identified as being, respectively, homogeneous unexsolved titanomagnetite (no pyrrhotite was seen), nearly pure magnetite subdivided by exsolved ilmenite lamellae, and hematite.

To date, no new components of NRM have been revealed by thermal cleaning, nor has resolution of superimposed components been much improved compared to AF cleaning.

To test for overlapping blocking temperatures of group 1 and group 2 NRM's, single-component samples carrying one NRM or the other were first compared. For the Shelley Lake granite at least, overlap was considerable and all three minerals were carriers of either component.

A second approach to the same question is illustrated in Fig. 5. In these Shelley Lake samples, the magnetization swung from a group 2 toward a group 1 direction on both AF and thermal demagnetization. Curiously enough, both normal and reversed group 1 components evidently coexist, the normal one being isolated by AF and the reversed one by thermal cleaning.

The group 1 component has the higher blocking temperatures in both samples, but the group 2 component possesses some blocking temperatures >600°C, since a stable group 1 endpoint has not

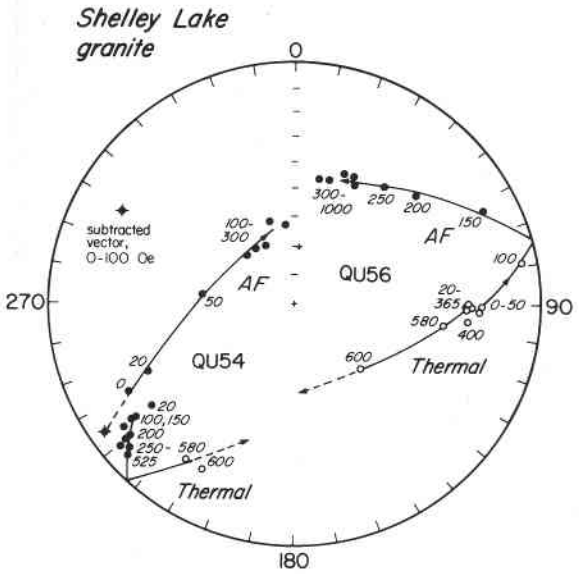


FIG. 5. Examples of multicomponent magnetizations in the Shelley Lake granite. The magnetization vector swings from an initial group 2 direction to a normal group 1 direction upon AF cleaning (numbers are AF's in oersted peak) and toward a reversed group 1 direction with thermal cleaning (numbers are temperatures in °C). Equal area projection.

been reached by 600°C. The test for relative ages is thereby invalidated. Even if group 1 blocking temperatures should prove to lie entirely above those of the group 2 component, this would merely indicate that hematite is the sole group 1 carrier. It does not follow that group 1 NRM is thermal and older.

Both group 1 and group 2 NRM's are likely largely chemical or thermochemical remanences. Nothing can be deduced from thermal demagnetization about their relative ages.

Paleopoles and Absolute Ages of the NRM Components

Paleomagnetic poles together with their ovals of confidence are calculated in Tables 2 and 3 and plotted for comparison with Tracks 3B, 4A, and 6 (Irving and McGlynn 1976; Irving and Naldrett 1977) of the North American APWP in Fig. 6. As noted earlier, only the group 1 results for the Shelley Lake and Burchell Lake granites and the Poohbah Lake complex (poles 1, 4, and 11 at the left of Fig. 6*a*) are of sufficient quality to serve in *defining* the APWP. The purpose of Fig. 6 is rather to use the APWP to date the group 1 and 2 poles.

The Poohbah Lake complex has been dated by Mitchell (1976) using K/Ar. He finds 2706 ± 23 Ma for the cooling age, with a later thermal event at 2556 ± 36 Ma. Berger and York (1979, this issue) have applied step-heating $^{40}Ar/^{39}Ar$ dating to the

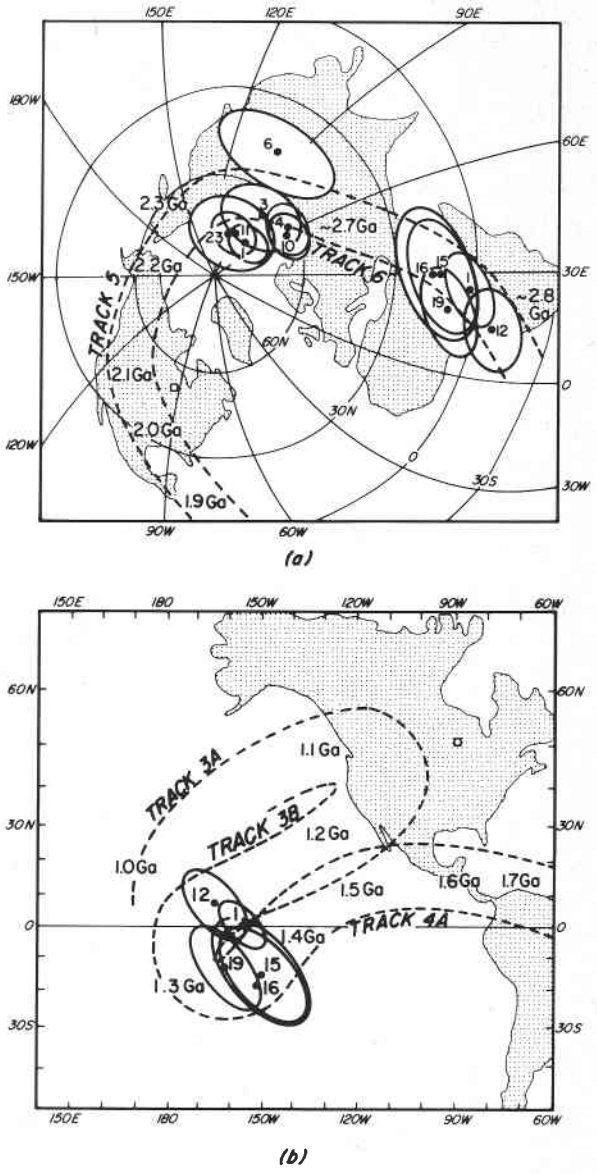


FIG. 6. Two segments of the Precambrian APWP for North America, redrawn after Irving and Naldrett (1977) and Irving and McGlynn (1976). Approximate time calibrations are given for the tracks but reference poles have been omitted. An important tie point, the Matachewan diabase pole, falls at the point marked 2.7 Ga in (*a*). Paleopoles shown, with their ovals of 95% confidence, are those of formations studied in this paper (Tables 2, 3), numbered as in Table 1. Group 1 poles are *about* 2.6 Ga old. Group 2 poles are either about 2.8 Ga or 1.35–1.25 Ga old.

Shelley Lake granite. They report a biotite plateau age of 2595 ± 14 Ma, with a thermal event recorded by potassium feldspars at about 1100 Ma.

The Poohbah Lake complex has only a group 1 NRM. In view of the discussion of the previous section, the group 1 NRM, for the Shelley Lake

granite anyway, is unlikely to be a primary TRM dating from initial cooling. Therefore, the group 1 NRM is probably about 2600 Ma in age, making it a metamorphic overprint associated with the cooling phase of the Kenoran orogeny. This inferred age agrees well with the approximate age calibration of Track 6 (Fig. 6*a*), whose tie point in the neighbourhood of the group 1 poles is the 2690 Ma (Gates and Hurley 1973) Matachewan diabase.

Assigning an age to the group 2 NRM is a more difficult problem. Figs. 6 *a* and *b* illustrate the possible alternative models. In model 1, the poles fall along Track 6 and the age assignment is about 2800 Ma. In model 2, the poles lie at the older end of Track 3B, with ages of 1250–1350 Ma. If we knew the relative ages of groups 1 and 2 from thermal demagnetization, the choice would be easy, but the best we can do is assess probabilities.

Model 1 implies a late Archean but significantly (~200 Ma) pre-Kenoran age for group 2. Favouring this model is the fact that group 1 NRM is ubiquitous and well-grouped while group 2 NRM is spotty in occurrence and more scattered in direction. A specific comparison is between the Shelley Lake granite, which has a group 2 component, and the Poohbah Lake complex, which lacks it. The Poohbah intrusion cuts across the grain of the Quetico gneiss belt and seems to be younger than the steeply-tilted (MacKidd 1973) Shelley Lake body, which conforms to the grain and must have been intruded before the sediments were deformed.

On the other hand, if group 2 remanence is pre-Kenoran, we would expect to have to apply a structural correction to allow for the change in attitude resulting from the Kenoran orogeny. In the Shelley Lake remanence, the group 2 component happens to lie nearly along strike and "untilting" the body has little effect on magnetization direction, but this cannot be true in general. The group 2 directions, however, group without any structural corrections, implying a post-Kenoran origin.

Model 2 implies a late Proterozoic remagnetization. Local tectonic events cannot be responsible because rocks of all belts have a common group 2 imprint. K/Ar ages in potassium feldspars from the Shelley Lake granite have been partially reset ca. 1100 Ma (Berger and York 1979, this issue). It would seem most natural to associate this resetting with the thermal effects of rifting and volcanism in the Keweenawan basin immediately to the southeast. The observed minima in $^{40}Ar/^{39}Ar$ spectra of potassium feldspars are, however, explicable if the heating event actually occurred about −1250 Ma (G. W. Berger, personal communication, 1978). A Mackenzie age is favoured by the magnetic data. According to Fig. 6*b*, the magnetic overprinting predates the onset of Keweenawan igneous activity by at least 100 Ma. Furthermore, the overprinting increases with distance from the Keweenawan rift.

The possibility of a regular progression of mean group 2 magnetization directions from one structural belt to the next was noted earlier. The same progression is seen in the corresponding paleopoles (Fig. 6). If the trend is real, there are two possibilities: (1) group 2 NRM was acquired at the same time in all belts, but the belts were not then in their present relative positions or (2) group 2 NRM was acquired at slightly different times in different belts, and the trend reflects drift of the Laurentian protocontinent during this time interval.

Radiometric dating does not have the precision for a direct test of simultaneity. Geometrical constraints on the first possibility will be considered in the section that follows. The second possibility can be ruled out if the sequence of poles obviously cuts across the trend of the APWP. (A well-known example is the pole for the Dogrib dikes of the Slave Province (McGlynn and Irving 1975) which is displaced from the contemporaneous Matachewan diabase pole in a direction normal to the APWP.)

Whether model 1 or model 2 holds, however, the group 2 poles trend along the APWP. The age progression is (from oldest to youngest) Shebandowan to Quetico to Wabigoon according to model 1 but the reverse according to model 2 . Such a trend might express, for example, burial metamorphism of the region, with uplift either beginning later or continuing longer to the northwest in model 1 or to the southeast in model 2.

Paleolatitude, Paleorotation, and Drift

Table 4 lists calculated paleolatitudes and paleoazimuths of selected formations at the times group 1 and group 2 NRM's were imprinted. Both models for the age of group 2 NRM are considered. The formations selected are those that record the extremes of continental drift, not those with the most reliable results. Paleolatitude λ_p is related to magnetic inclination I by $\tan \lambda_p = \frac{1}{2} \tan I$ and the angle between paleonorth and present-day north is the declination D (or $180° - D$ for reverse magnetizations).

Figure 7*a* shows that the Burchell Lake granite was located about 35°N when it acquired its group 1 magnetization (the Shelley Lake and Watcomb granites have similar paleolatitudes), the Poohbah complex (and the Huronian granite–paragneiss) were about 20°N, and the Sapawe iron formation *may* have been equatorial. There is negligible paleorotation.

TABLE 4. Paleolatitudes and paleoazimuths of the continent or continental nucleus containing the Superior Geotraverse area at the times various formations became magnetized

Formation	Paleolatitude	Rotation relative to present continent
Time of group 1 magnetizations (~2.6 Ga, Track 6)		
(4) Poohbah Lake complex	22.2N	12.3° counterclockwise
(6) Sapawe iron formation	6.8N	8.5° clockwise
(11) Burchell Lake granite	34.8N	No rotation
(23) Watcomb granite	36.1N	3.3° counterclockwise
Time of group 2 magnetizations (~2.8 Ga, Track 6)		
(1) Shelley Lake granite	16.2S	70.5° counterclockwise
(12) Shebandowan gabbros, volcanics	15.2S	82.9° counterclockwise
(15) Gulliver greenstone	7.4S	57.3° counterclockwise
Time of group 2 magnetizations (~1.25 Ga, Track 3B)		
(1) Shelley Lake granite	16.2N	109.5° clockwise
(12) Shebandowan gabbros, volcanics	15.2N	97.1° clockwise
(15) Gulliver greenstone	7.4N	122.7° clockwise

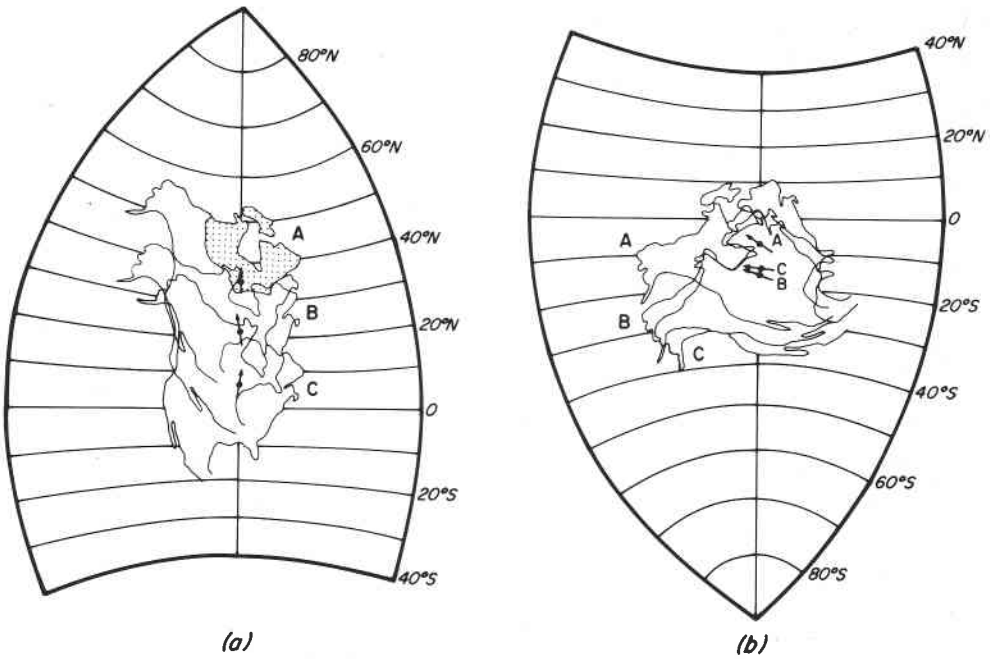


FIG. 7. Suggested drift of the Laurentian protocontinent (the presently exposed part is shaded) (*a*) at the close of the Kenoran orogeny, about −2.6 Ga, and (*b*) about −2.8 Ga, the inferred age of group 2 NRM. In (*a*), A, B, and C represent positions of Laurentia at the times group 1 NRM was acquired by the Burchell Lake granite, the Poohbah Lake complex, and the Sapawe iron formation respectively. In (*b*), A, B, and C are the positions when group 2 NRM was acquired by the Gulliver greenstone (Wabigoon), the Shelley Lake granite (Quetico), and the Shebandowan mafic rocks (Shebandowan) respectively. Arrows indicate paleonorth in each case.

The time calibration of Track 6 (Fig. 6*a*) implies that the time interval for drift could be as wide as 2.6–2.4 Ga and that the Poohbah Lake complex was overprinted earlier than the Shelley Lake or Burchell Lake intrusions. The Kenoran orogeny accompanied a period of perhaps rapid northward drift of Laurentia, but whether collision with another protocontinent caused the orogeny, we cannot say. There is no evidence that relative drift of tectonic belts within Laurentia was involved, causally or incidentally, in the orogeny, although the motion traced in Fig. 7*a* is in the right general direction to close paleooceans between the generally east–west-trending belts.

Figure 7*b* illustrates the overall and relative paleopositions of some Shebandowan, Quetico, and Wabigoon belt formations about 2.8 Ga ago. (If model 2 is accepted, the paleopositions 1.25–1.35 Ga ago can be obtained by inverting Fig. 7*b* and redefining the south pole as the north pole.) Laurentia was nearly equatorial and rotated almost 90°. If the implied relative displacements of the belts are real, they involved substantial differential rotation, small differences in paleolatitude, and unknown differences in paleolongitude.

Since paleonorth is present-day east, circumstances are unfavourable for detecting opening or closing of oceans between presently east–west (paleo-north–south) belts: such motions would have changed only paleolongitude. Paleolatitude changes would record paleo-north–south motion, i.e., shear between the belts. The only significant paleolatitude difference in Fig. 7*b* is between the Gulliver greenstone result and those for the other formations. If real, the difference implies that the Wabigoon belt at the time was > 500 km east of the Shebandowan and Quetico belts, reaching its present location by westward (sinistral) shear, counterclockwise rotation of ~20°, and an unknown amount of convergence.

The geological evidence, however, favours dextral rather than sinistral shear on the Quetico fault (Mackasey *et al.* 1974), with displacements of at most 200 km. A fundamental objection is that large-scale movements, eastward or westward, would degroup the older group 1 poles, unless the movements represented the second half of a Wilson cycle and reassembled the subprovinces in their Kenoran configuration.

The more likely explanation is that these relative paleopositions record successive locations of the entire Laurentian continent at different times or that they are an artifact resulting from noisy data.

Conclusions

The main conclusions of this study are the following:

(1) As a result of greenschist-facies metamorphism, volcanic rocks from the southern Superior Geotraverse region are weakly and rather unstably magnetized compared to modern oceanic basalts. Mafic intrusions are more strongly and stably magnetized than their volcanic equivalents, felsic and intrusions sometimes much more so.

(2) No simple criteria were discovered relating magnetic intensity and stability to either rock type or degree of metamorphism.

(3) Two stable ancient NRM's are preserved in many different types of rocks from the area. One is a ubiquitous metamorphic overprint dating from the Kenoran orogeny. The other could be Archean, either a primary remanence or a pre-Kenoran overprint, but it is more likely an overprint acquired during late Proterozoic time.

(4) Thermal demagnetization failed to determine the relative ages of the two NRM's. Both components are probably chemical or thermochemical overprints.

(5) Paleopoles for the two NRM's when compared to the Precambrian APWP for North America suggest an age of ~2600 Ma for the Kenoran orogeny, in agreement with two radiometric dates from the area, and a mild metamorphic event about 1250–1350 Ma ago, probably not related to Keweenawan igneous activity.

(6) At the time of the Kenoran orogeny, the Wabigoon, Shebandowan, and Quetico belts already formed a single unit within Laurentia. During the orogeny, Laurentia was in its present orientation and the geotraverse area was located between 20 and 35°N.

(7) At the time the second characteristic NRM formed, either 2800 Ma or, more likely 1250–1350 Ma ago, Laurentia was equatorial and rotated about 90° from its present orientation.

(8) The Wabigoon subprovince *may* have been displaced from the Quetico and Shebandowan belts about 1350–1250 Ma ago, having reached its present position by major sinistral shear and an unknown amount of convergent motion.

(9) Alternatively, 1350–1250 Ma ago there may have been no relative displacement of belts but the second NRM may have been acquired later in the Shebandowan and Quetico belts than in the Wabigoon belt.

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